NETWORK DEFENSE INTELLIGENCE EXERCISE

Independent Research Execution • Technical Feasibility • Adversarial Coverage • Experiments • Economics • Funding

Document ID: OAL-NDIE-RESEARCH-EXEC-CHATGPT-001 | Version: 1.0 | Date: 5 November 2026

Research mandate: OAL-RESEARCH-NDIE-001 v1.1 | Primary scenario: OAL-DOSSIER-CANON-002 v1.0.0 (pending ONESTAMP)

Audience: ChatGPT · Claude · Gemini · Copilot · Perplexity · Grok · OAL GitHub archive

Status: Independent desk research and proposed experiments; no simulations, attack trials, pilot deployments, agency partnerships, or cost quotations were performed or obtained.

Executive decision memorandum

Decision: CONDITIONAL GO for a small, independently evaluated, isolated research study; NO-GO for a national-scale or production implementation on current evidence. Confidence: moderate in the narrow recommendation, low in claims of national-scale feasibility. A convincing fictional design is not evidence of a deployable system.

Goal: determine whether selectively deployed decoy environments plus controlled response orchestration reduce unauthorized reachability of real assets, improve time-to-containment or yield useful incident intelligence compared with an equally resourced conventional security baseline. The defensive goal is protecting mission assets—not maximizing the number of sessions drawn into decoys.

Most consequential finding: an adversary can pursue the protected data or mission outcome through trusted identities, existing footholds, suppliers, workflows, physical interfaces, or an unmonitored control plane. Those threats may never touch the deception surface. Therefore any evaluation must measure bypass as seriously as deception engagement.

Technical finding: individual elements—honeynets, dynamic reconfiguration, cyber ranges, automated vulnerability remediation, and isolation—have public precedents. The reviewed sources do not establish transparent migration of arbitrary authenticated sessions into a procedural national replica, optimal 3-second rekeying, omniscient end-to-end attribution, or total capture of active attackers [S01–S10].

Economic finding: this report supplies transparent scenario budgets, not vendor quotations or independently validated program prices. An initial 3-month study is modeled at ~$131,000 base case; a 6-month controlled proof of concept at ~$588,000; a 12-month range evaluation at ~$2.31 million; and a 24-month multi-organization synthetic pilot at ~$8.12 million. These are planning illustrations contingent on staffing and scope, not funding requests.

Funding finding: NSF SaTC 2.0, certain DARPA BAAs, and eligible SBIR/STTR pathways merit assessment, but agency fit, applicant eligibility, deadlines and actual award availability must be rechecked when submitting [S11–S14].

1. Source audit and epistemic boundaries

The source dossier labels the scenario “fixed canon” while simultaneously stating that no item is fully ratified until an author-applied ONESTAMP. This report therefore uses “stated canon constraint” for the protected story sequence, “research hypothesis” for technical claims, and “proposed amendment” for design changes. DOCX/PDF versions appear to represent the same source; no substantive version-difference was established by a structured line-by-line parity test.

Issue / source location	Effect on research	Action
Approval paradox (dossier cover; §2)	Fixed narrative versus pending ratification	Do not mark as approved; preserve change requests.
Five reviewing AIs versus six-member distribution (§1; §6.2)	Provenance not independently verified	List actual submissions and receipt dates; do not treat agreement as independent replication.
Five-tier framework with Tier 0 and Tier 5; critical Tier 0/Tier 1 (§6.1)	Control selection and transfer classification ambiguous	Obtain authoritative tier definitions before implementation.
Internal question and failure-case cross-references (§4.4; §8–10)	Potential traceability errors	Create corrected issue register, without altering source.
“All attackers captured” (§2; §4.4)	Unknown denominator makes universal proof impossible	Define bounded test population and independent ground truth.
3-second pulse (§2; §4.3)	Narrative constant not measured optimum	Treat as one treatment arm, not a standard.

Source attribution note: Prior AI prompt files influenced the mandated questions; they are not independent experimental sources. This is ChatGPT’s independent synthesis of the attached dossier and publicly retrieved references. No other AI’s research report has been reviewed or validated for this execution.

2. Purpose, importance and counterfactual

Mission problem: defenders often have incomplete visibility, uncertain attribution and limited response time; deception can provide controlled observation and potentially useful high-signal encounters. MITRE D3FEND explicitly catalogs decoy environments and decoy network resources; NIST SP 800-160 v2 r1 identifies deception/unpredictability as resilience techniques while warning about operational uncertainty [S01–S03].

Null hypothesis H0: after equal expenditure and comparable defensive baseline controls, adaptive deception produces no material improvement in real-asset protection, time-to-containment, independently validated intelligence quality, or operator workload. Rejection requires preregistered thresholds and repeated randomized trials, not attractive visualizations or raw honeypot contact counts.

Counterfactual: fund stronger identity governance, segmentation, least privilege, endpoint monitoring, software assurance, backups and recovery without adaptive decoys. NIST SP 800-207 emphasizes protection of resources and identities rather than reliance on network location [S04]. A decoy system should be additive and removable, with the independent protective controls still functioning.

Minimum useful research outcome: a replicable measurement of whether fixed or adaptive decoys measurably help, where they fail, and at what cost. Minimum useful product: owned, isolated decoy assets with verified no-path-to-production boundaries, high-quality logs, a human-governed alert workflow, and safe rollback. Neither requires duplicating an entire nation.

3. Capability decomposition and feasibility assessment

Capability	Evidence classification	Boundary / unresolved test
Low/medium-interaction deceptive services	Demonstrated in tools [S02,S05]	Protocol realism and detectability vary.
Connected or standalone honeynets	Defined, deployed in practice [S01,S05]	Isolation must be tested; connection is not capture.
Moving-target/network mutation	Research-stage and partly demonstrated [S06]	Latency, resource overhead and session stability understudied.
Synthetic workload generation	Plausible at bounded scale	External truth and physical process fidelity remain hard.
Selective redirect of new flows	Plausible under owned ingress	Requires control of the entry point, routing and policy.

Invisible migration of arbitrary established encrypted sessions	Unsupported as a general claim	Auth, state, transport and app semantics prevent universal migration.
Complete attack-path mapping and attribution	Partial, inference-dependent	Unobserved routes and intermediaries prevent universal certainty.
Every active attacker contained; real environment fully secured	Not independently verifiable at open-world scale	Only bounded known test populations permit measured coverage.
Three-second nation-scale key/topology cycling	Speculative	No reviewed validation of scale, safety or optimality.
Secure package transfer and partner rollout	Ordinary engineering mechanisms exist	Provenance, assurance, authority and rollback needed.

4. The neutralization-to-migration seam

A defensible real-world interpretation has three separate operations: (a) enforce a real-asset denial policy with independent default controls; (b) offer an authorized controlled decoy path where a new or selected synthetic session can begin; and (c) observe and stop the decoy interaction without making unsupported claims about the adversary. Preventing real access before redirection may cause sessions to fail or users to notice; leaving access active until redirection creates an exposure window. Both outcomes must be measured.

Transport/network redirection is not equivalent to transplanting authenticated application state. Session-bound credentials, TLS identity, application consistency, external systems and non-replayable effects are major constraints. Do not attempt or claim seamless arbitrary production-session migration. Test only explicitly instrumented synthetic sessions and record redirection eligibility, failed transfers, interruption and preservation of the protected resource boundary.

A required safety invariant is: regardless of decoy controller state, the test attacker cannot obtain an unauthorized route to the protected synthetic “real” asset. The invariant must be enforced by a separately administered control in a disconnected range, not by the deception classifier alone.

5. Precedent evidence review: what exists, what worked, what remains unproven

The following examples intentionally include frameworks, exercises, competitions, deployed tooling and systematic research reviews. They are not interchangeable: an exercise finding is not a product efficacy estimate, and a framework is not an experiment.

Reference / type	Observed contribution	Caveat and transferable lesson
MITRE D3FEND decoy taxonomy [S01]	Defines connected/integrated/standalone honeynets and deceptive resources.	Taxonomy, not effectiveness trial; use for clear capability vocabulary.
Cowrie project [S05]	Operational SSH/Telnet emulation and captured session logging; optional proxy/high-interaction paths.	Narrow protocol coverage; useful inexpensive bounded test instrument.
NIST cyber-resiliency engineering [S03]	Recognizes deception and unpredictability as defensive approaches.	Also warns of admin/user uncertainty; require usability measures.
CISA Cyber Storm IX, April 2024 [S07]	>2,200 participants; cloud/critical-infrastructure incident-response exercise.	After-action lessons concern coordination and cloud governance, not invisible national redirection.
NATO CCDCOE Locked Shields [S08]	Cross-national, realistic critical-service exercise involving legal and communications concerns.	Large exercise ≠ proof of synthetic-network capture; federation and human processes matter.
DARPA Cyber Grand Challenge 2016 [S09]	Autonomous competition for vulnerability discovery, exploitation and patching.	Artificial challenge setting; no claim of universal enterprise protection.
DARPA AI Cyber Challenge 2025 [S10]	Autonomous systems identified 54 synthetic vulnerabilities and patched 43 across 63 challenges.	Software repair benchmark; not honeypot or national defensive-network proof.
Honeypot fingerprinting review (2026)	Surveys identifiable protocol/state/timing	Negative/limiting evidence: clever

[S06]	artifacts across deceptive systems.	attackers may recognize decoys.
MTD-in-SDN review (2026) [S15]	97 studies reviewed; latency/throughput/resource-overhead measurement gaps.	Negative/limiting evidence: published experiments often fail to evaluate operational costs.
LLM-honeypot controlled study (2026) [S16]	Controlled proof of concept reports exploratory detection/analysis.	Negative/limiting evidence: 52 observations from one run, limited external validity.

Research gap: no source in this set reports the full proposed OAL chain at national scale. In particular, research on honeypot detection/avoidance and weak operational benchmarking actively challenges assumptions about undetectability and safe rapid cycling. Quantitative comparisons across the above programs are not warranted because their datasets and outcomes differ.

6. Asset-centered adversarial bypass coverage

Threat-model unit: a protected mission asset and its authorized access paths. “Bypass” means the adversary reaches the intended effect without passing a monitored decoy gate; it is not necessarily evasion of a detection signature. The matrix below is deliberately non-operational.

Bypass class	Why deception may not engage	Independent safeguard / safe evaluation
Legitimate identity or delegated access	Access looks authorized at network layer	Identity-aware authorization; synthetic role/privilege misuse trials.
Supplier or update trust	Trusted artifact reaches asset via normal process	Signed/provenance-checked releases; simulated malicious package rejection.
Preexisting presence	Actor already operates inside boundary	Asset-level monitoring; synthetic resident-user scenarios.
Out-of-band / unmanaged / physical	No traffic reaches decoy-controlled gateway	Asset inventory, facility/process controls; tabletop and disconnected test fixtures.
SaaS data copies / third parties	Data resides beyond owned decoy network	Contractual controls, data governance, cloud audit; mock external workflows.
Control-plane corruption	Defender loses trusted evidence or policy control	Independent reference policy, attestations; failover/rollback injection.
Telemetry poisoning	Misleading samples contaminate automatic responses	Quarantine untrusted logs; blind synthetic poisoned-data trials.
No engagement / patient actor	Attacker declines all bait	Measure bypass coverage and independent loss prevention.
Disruption-only objective	Damage can precede or avoid session capture	Availability controls, rate limiting and recovery; synthetic load tests.
Multi-actor distractions	Decoy engagement diverts scarce analysts	Workload caps and prioritization; randomized analyst exercises.

Residual risk: no set of network decoys can guarantee defense against all physical, identity, supply-chain or organizational pathways. Across repeated episodes, attacker awareness can decrease engagement while defender costs persist. Unknown blind spots must remain explicit, never counted as protected just because there were no decoy alerts.

7. Defender–adversary evolution over six encounters

Encounter	Adversary posture	Defender test / preferred response
0	Unaware of deception	Measure initial engagement and baseline protection.
1	Aware deception exists	Test deterrence vs movement to other surfaces.
2	Suspicious; evaluates consistency	Test realism, false positive burden, timing signatures.
3	Recognizes recurring patterns	Compare static, randomized and event-

		driven refresh.
4	Avoids decoys; leverages trust boundaries	Recenter on identity, supply chain, asset controls; measure zero-engagement bypass.
5	Attempts to influence classification or learning	Freeze retraining, quarantine evidence, independent confirmation and rollback.

Learning policy: offline reviewed retraining only; preserve held-out attack episodes and immutable logs; version prompts/models/policies; audit drift, alert fatigue and operator overtrust. NIST AI 100-2e2025 categorizes relevant adversarial ML threats [S17]. Trigger pause or rollback on high false-positive rates, unexplained access-control changes, provenance failures or unexplained real-asset reachability.

8. Three-second cycle: factorial experiment

Hypothesis: a 3-second fixed refresh could increase uncertainty for an observing actor, but may also become a recognizable signature, disrupt sessions, increase resource consumption and impair forensics. No optimality claim is justified. Independently vary both what changes and when: (a) static, (b) fixed 3-second, (c) random interval matched for average refresh, (d) risk-triggered interval. Keep each treatment's compute allocation and admission policy constant, measuring network availability and defender observability.

Variables: refresh interval distribution; service/persona presentation; synthetic identity; topology within the lab. Avoid automatic rotation of real production cryptographic keys or broad network routes. Outcomes: completion of legitimate synthetic sessions, failure/latency percentiles, resource cost, recognition by assigned red teams, event correlation and recovery time. A fixed cycle is disfavored if cost, visibility or reliability worsens without a significant asset-protection gain.

9. Seven controlled experiments and progression gates

Test	Design / comparator	Outcome and hard stop
E1 Baseline increment	Same zero-trust/segmentation controls; + static decoy or adaptive decoy	Unauthorized reachability, time to isolate; halt if real-asset boundary fails.
E2 Bypass population	Synthetic scenarios that do not touch decoy vs gate-crossing scenarios	Per-class coverage; no alert must not be scored as success.
E3 Cycle cadence	Static, fixed 3s, random, event-triggered	Latency, breakage, compute, recognition, reconstructability.
E4 Provisioning	Predeployed decoys vs on-demand procedural rendering	Startup lag, fidelity, cost, unsafe error rate.
E5 Learning	Withheld test cohort over six successive encounters	Engagement decline, model poisoning resistance, change in bypass.
E6 Innocent traffic	Known benign monitors, users and misconfigurations	False diversion, human correction, workflow harm; stop if preset disruption exceeded.
E7 Controller and update failure	Deliberate synthetic control failure; corrupted *dummy* package	Independent deny policy holds, safe rollback, provenance verification.

Proposed initial power method: pilot each condition to estimate variance; pre-register a minimally important difference (e.g., 20% lower median isolation time without more than 1 percentage point increase in false diversion) as a *proposed* threshold, then compute sample size before confirmatory runs. Randomize scenario order; use separate seeds and blind analysts where feasible. The 20% and 1-point values are design proposals, not observed results.

Metrics include recall and precision against a fully known synthetic ground truth, false diversion/1,000 benign sessions, unauthorized reachability/attack opportunity, time-to-isolation, p95 latency, decoy recognition rate, compute-hours per protected asset, actionable forensic yield, rollback duration, and analyst minutes per case.

Report confidence intervals and all failures; prohibit “100% capture” outside the enumerated experimental population.

Progression gates: G0 define assets/legal owner/metrics; G1 evidence-based justification; G2 isolated sandbox security review; G3 independent replication and safe rollback before limited external pilot; G4 verified governance/financial justification for any wider scope. OT/ICS requires separately designed disconnected emulation and subject-matter approval. No national or third-party live trial is authorized.

10. Budget modeling: auditable, illustrative USD scenarios

Method: loaded labor = FTE × months/12 × assumed annual loaded compensation. Assumed fully loaded blended rates: $190,000 (study); $205,000 (PoC); $220,000 (range); $225,000 (multi-organization). These are model inputs, NOT BLS salary estimates. BLS reports 2025 median information-security analyst wage of $129,180 (salary only) [S18]; multipliers incorporate employer costs, specialist mix, and contractor overhead and are unverified. Direct nonlabor includes computing, tools, evaluation, governance, and contracted facilities; contingency is 20% of labor + direct costs. No quoted cloud pricing was obtained.

Stage	Term / staffing	Labor	Other direct	20% reserve	Base estimate
Desk feasibility	3 mo / 1.5 FTE	$71,250	$38,000	$21,850	$131,100
Isolated proof of concept	6 mo / 4 FTE	$410,000	$80,000	$98,000	$588,000
Controlled cyber range	12 mo / 7 FTE	$1,540,000	$385,000	$385,000	$2,310,000
Synthetic multi-org pilot	24 mo / 12 FTE	$5,400,000	$1,370,000	$1,354,000	$8,124,000

Illustrative uncertainty bands: low = 75% of base; high = 150% of base. These are transparent sensitivity envelopes, not statistically estimated prediction intervals or procurement bids. For the PoC this implies roughly $397,000–$794,000; for the range ~$1.59m–$3.19m; for the multi-organization stage ~$5.44m–$10.89m. Sensitivity is greatest to staffing duration, fidelity, partner integration, realistic OT simulation and independent evaluation.

Equal-budget alternative for the six-month PoC: allocate the same ~$588k to documented identity/least-privilege experiments, segmentation testing, endpoint/monitoring improvements, incident-response rehearsal and recovery engineering. If this alternative has equal or better real-asset metrics, do not scale the deception prototype. Hypothetical U.S.-wide deployment costs are deliberately NOT estimated: organizational boundaries, legal permissions, ownership, telemetry scope and safety requirements have not been specified.

11. Funding and partnership pathways (verified program descriptions, not awards)

Path / current evidence	Fit	Uncertainty / eligibility gate
NSF SaTC 2.0, NSF 25-515 [S11]	Strong for university-led cyber trust, safety, measurement; RES up to $1.2m/4 years, SEED up to $300k/2 years subject to terms.	Check eligible submitting institution/PI; target dates listed for Jan/Sept 2027; review updates before applying.
DARPA office-wide BAAs [S12]	Potential for novel high-risk engineering if problem is truly transformative.	Competitive fit; no implied contract; inspect live BAA, security, data rights, program needs.
DOE SBIR/STTR [S13]	Small-business commercialization of relevant energy security technology if topic permits.	Only eligible U.S. for-profit small businesses; open topics and stage eligibility require confirmation.
NIST SBIR [S14]	Possibly relevant technical proof of concept under matching annual topic.	No project entitlement; annual NOFO, small-business requirements.
DHS S&T / CISA	Mission alignment on resilience and exercises; possible future partnerships.	No currently open matching award or eligibility confirmed in this review.

University / lab / private co-funding	In-kind cyber range, independent evaluation, practitioner access.	Requires written letters, data governance and conflict management.

Financing sequence: begin with a 2–3 page neutral concept note, small measurable hypothesis, institutional PI/partner, documented technical risk, and modest budget. Avoid pitching “national-scale capture” as a verified capability. Current opportunity status must be revalidated at application time; publication on a program website is not an award or endorsement.

12. Governance, transfer artifact and safety case

Real-world authority is distributed across independently owned networks. A centralized orchestrator must never be assumed to have universal visibility or legal control. Prefer federated administration with per-owner authorization, scopes, time-bounded capability grants, role separation, independent kill switches and auditable decisions. Avoid sole operator power over multiple organizations.

The partner-agency update package is a high-consequence supply-chain surface. Required controls: component and build provenance; signed release manifest and SBOM; independent receiving-party validation; constrained configuration; separated signing/key custody; staged release; revocation; documented rollback; tamper-evident logs; limited handling of collected intelligence; and retention/deletion schedules. Avoid storing operational secrets in public GitHub. Classify the artifact only after resolving source taxonomy ambiguity.

Privacy and legal questions: who owns intercepted communications; what is consent for external contributors; how to minimize personal data; which retention and sharing rules apply; what approvals govern government or critical-infrastructure tests; and how to prevent diversion of legitimate users. This report is not legal advice; counsel and host organizations must decide before real-world trials.

13. Strongest credible case against funding

- The incremental protective value may be small once identity-based access, segmentation, detection and response are adequately funded.
- Attackers may shift to privileged or indirect routes that never engage decoys; observed honeypot success could create false confidence.
- Highly realistic replicas may reproduce sensitive system information, increase attack surface and cost more to operate than the assets defended.
- Fast cyclic mutation may harm session reliability and become a detectable characteristic rather than a deterrent.
- Automated classification and retraining can increase false positives, contaminate evidence and overload operators.
- Federated authority and jurisdictional reality make the national version operationally and legally ill-defined.
- The post-exercise deployment artifact could introduce a new consequential supply-chain failure mode.

A rational no-go result arises when benefit is not distinguishable from baseline after adequately powered tests, when bypass classes dominate mission risk, when operator burden or benign disruption increases, or when safe independent auditing and rollback cannot be demonstrated.

14. Conditional 30 / 60 / 90-day research plan

Window	Work package	Go/no-go deliverable
Days 1–30	Source audit; scope and asset register; 10-source evidence ledger; stakeholder/PI mapping; protocol design.	Approve measurable H0/H1, synthetic scope, independence and budget; otherwise defer.

Days 31–60	Implement offline synthetic range for E1–E4; instrument baseline and static decoy; measure handoff and 3s variants.	Initial feasibility/overhead report; stop for invariant breach.
Days 61–90	Run E5–E7; red-team study with blinded reviewers; compute cost and baseline comparison; independent replay.	Decision: redesign, narrow research, terminate, or seek formal pilot funding.

15. Proposed design revisions to fictional canon (not approved)

Retain the detection→mapping→neutralization→synthetic diversion→three-second pulse→black-line collapse→transfer sequence as a storytelling constraint pending ONESTAMP. Clarify that the system has partial visibility in the underlying mechanics; “all attackers” is only what the Architect reasonably asserts or observes; an independent actor may remain beyond control; the three-second rhythm might be visual/decoy state rather than real Internet routing changes; and the transferred package poses a governance dilemma. These are author-facing options, not silent changes.

Suggested failure scene: one black line never collapses, not because the adversary defeated the decoy, but because it represents a trusted workflow not covered by the deceptive layer. The defenders discover the discrepancy between a clean network display and an unresolved asset-level consequence. This introduces limits without negating the established visual motif.

16. Evidence ledger, search limitations and confidence

Research method: targeted public web searches conducted 5 Nov 2026 for NIST, MITRE, DARPA, CISA, CCDCOE, NSF, DOE, NIST SBIR, official tool documentation and recent scholarly reviews. Priority: primary standards and official after-action reports; secondary systematic reviews used to identify gaps. This was a curated desk review, not a reproducible systematic literature review; no full-text meta-analysis, raw benchmark datasets, direct subject interviews, or independent replication. Agency program availability could change after this research date. One CISA PDF screenshot request was blocked by the source server, but accessible parsed after-action text and official CISA prose were used.

Claim confidence: HIGH—MITRE taxonomic decoy definitions, NIST resource-centric zero trust, published program/competition descriptions. MODERATE—limited value and risks of deception/MTD derived from reviews, because studies differ greatly. LOW—expected comparative performance, 3-second benefit, predicted adversary decisions, national feasibility and budget precision; these remain hypotheses or illustrative planning inputs.

Independent research record: the source documents and current web pages were evaluated by this model alone; no outside AI team outputs or GitHub updates are represented as independently verified research findings. Model-generated analyses are not experimental observations.

17. Researcher-to-team exchange: three questions

1. What independently measurable protective benefit remains if a patient adversary never touches a decoy, and how should that affect the project’s go/no-go decision?
2. At which stage does greater central coordination become a larger systemic or governance risk than the original intrusion threat, and what federated design can reduce that risk?
3. If a conventional, equally funded security upgrade equals or exceeds the deception treatment, which real-world modules should be retired without changing the creative scenario?

Appendix A — Source register (publicly located as of 5 Nov 2026)

S01 MITRE D3FEND, Deceive / Decoy Environment

https://d3fend.mitre.org/tactic/d3f%3ADeceive/

Use / caution: Definitions, not efficacy measurement.

S02 MITRE D3FEND, Decoy Network Resource

https://d3fend.mitre.org/technique/d3f%3ADecoyNetworkResource/

Use / caution: Operational considerations; not a trial.

S03 NIST SP 800-160 v2 r1, Developing Cyber-Resilient Systems (2021)

https://csrc.nist.gov/pubs/sp/800/160/v2/r1/final

Use / caution: Deception/unpredictability and resilience engineering.

S04 NIST SP 800-207, Zero Trust Architecture (2020)

https://csrc.nist.gov/pubs/sp/800/207/final

Use / caution: Asset, subject, and resource-oriented authorization.

S05 Cowrie open-source honeypot documentation

https://github.com/cowrie/cowrie

Use / caution: Technical project evidence, not independent deployment study.

S06 Chaudhry et al., Honeypot Fingerprinting Review (2026)

https://doi.org/10.3390/fi18040190

Use / caution: Secondary research: fingerprinting limitations.

S07 CISA, Cyber Storm IX After-Action Report (2024)

https://www.cisa.gov/sites/default/files/2024-10/Cyber%20Storm%20IX%20After-Action%20Report%20v00%2020241001_508.pdf

Use / caution: Exercises coordination; source PDF access limits noted.

S08 NATO CCDCOE, Locked Shields exercise

https://ccdcoe.org/locked-shields/

Use / caution: Official exercise overview.

S09 DARPA, Cyber Grand Challenge Winners (2016)

https://www.darpa.mil/news/2016/cyber-grand-challenge-winners

Use / caution: Autonomous competition results.

S10 DARPA, AI Cyber Challenge results (Aug 2025)

https://www.darpa.mil/news/2025/aixcc-results

Use / caution: Automated code vulnerability/patch benchmark.

S11 NSF SaTC 2.0 solicitation 25-515

https://www.nsf.gov/funding/opportunities/nsf-satc-us-national-science-foundation-security-privacy-trust

Use / caution: Program fit, ceilings and target dates.

S12 DARPA office-wide BAAs

https://www.darpa.mil/research/opportunities/baa

Use / caution: General access and constraints; award not guaranteed.

S13 DOE SBIR/STTR program

https://www.energy.gov/technologycommercialization/doe-small-business-innovation-research-sbir-and-small-business

Use / caution: Small-business eligibility and current program.

S14 NIST SBIR program

https://www.nist.gov/tpo/small-business-innovation-research-program-sbir

Use / caution: Current program background and reauthorization.

S15 Moving Target Defense in SDN: Systematic Review (2026)

https://ieeexplore.ieee.org/abstract/document/11414228

Use / caution: Secondary review, evaluation gaps; publisher abstract.

S16 Integrated Honeypot and LLM Framework (2026)

https://www.mdpi.com/2624-800X/6/5/147

Use / caution: Single-run study and explicit limitations.

S17 NIST AI 100-2e2025, Adversarial Machine Learning

https://doi.org/10.6028/NIST.AI.100-2e2025

Use / caution: Adversarial learning terminology and risk framework.

S18 U.S. BLS, Information Security Analysts, May 2025 wage

https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm

Use / caution: Salary reference; not loaded contract rate.

Appendix B — Assumptions, falsifiers and records for next iteration

ID	Research assumption	Would change the conclusion
A01	A decoy can produce net incremental defensive utility	Equal-budget controls outperform deception on asset outcomes.
A02	At least some engagement can be safely diverted	Session-state trial shows unacceptable failure or exposure.
A03	Adaptive policies do not outweigh their overhead	Random/event-driven vs 3s loses on availability and forensics.
A04	Automated learning can be governed and audited	Poisoned-data tests cause uncontrolled policy changes.
A05	A bounded study is financially and legally viable	No qualified host or funding/authorization path exists.

Proposed next iteration review: record each collaborating model's point-by-point objections, source additions, updated estimates with provenance, counterexamples, and requested changes. A separate decision register—not this research report—should determine whether the prompt, scenario, or technical hypothesis is amended.